SQL Query Investigation Lab

Title

Investigating Employee Devices and Login Activity Using SQL

Overview

In this lab, I used SQL queries within the MariaDB shell to retrieve, analyze, and organize security-related data from the `organization` database. The investigation focused on employee devices requiring updates and login activity monitoring to identify potential unusual behavior.

This exercise demonstrates foundational cybersecurity data analysis skills using SQL, including querying databases, selecting relevant fields, and sorting event data for security investigations.

Objectives

- Retrieve employee device information from a database
- Identify device operating systems and patch status
- Investigate user login activity across locations
- Review login attempts outside expected working hours
- Organize login data chronologically using `ORDER BY`
- Strengthen SQL skills used in cybersecurity operations

Environment

Component	**Details**
Database	MariaDB
Database Name	Organization
Tables Used	machines, log_in_attempts
Operating Environment	Linux Bash Shell

Task 1 – Retrieve Employee Device Data

1. Retrieve All Device Information

SQL Query

SELECT *
FROM machines;

```
MariaDB [organization]> SELECT * FROM machines;
```

device_id	operating_system	email_client	OS_patch_date	employee_id
a184b775c707	OS 1	Email Client 1	2021-09-01	1156
a192b174c940	OS 2	Email Client 1	2021-06-01	1052
a305b818c708	OS 3	Email Client 2	2021-06-01	1182
a317b635c465	OS 1	Email Client 2	2021-03-01	1130
a320b137c219	OS 2	Email Client 2	2021-03-01	1000
a398b471c573	OS 3	Email Client 2	2021-12-01	0
a667b270c984	OS 1	Email Client 1	2021-03-01	1078
a821b452c176	OS 2	Email Client 2	2021-12-01	1104
a998b568c863	OS 3	Email Client 1	2021-12-01	1026
b157c491d493	OS 2	Email Client 1	2021-03-01	0
b239c825d303	OS 1	Email Client 1	2021-03-01	1001
b264c773d977	OS 2	Email Client 2	2021-03-01	1157
b265c937d713	OS 2	Email Client 1	2021-09-01	1131
b433c245d868	OS 1	Email Client 1	2021-06-01	1079
b551c837d758	OS 3	Email Client 1	2021-03-01	1105
b566c710d544	OS 1	Email Client 1	2021-06-01	1183
b806c503d354	OS 2	Email Client 1	2021-12-01	1027
b979c871d361	OS 2	Email Client 1	2021-03-01	1053
c116d593e558	OS 3	Email Client 1	2021-09-01	1002
c150d982e144	OS 2	Email Client 2	2021-06-01	1132
c185d679e493	OS 1	Email Client 2	2021-09-01	0
c406d877e950	OS 2	Email Client 1	2021-06-01	1158
c547d140e477	OS 2	Email Client 1	2021-03-01	1054
c568d742e974	OS 2	Email Client 2	2021-09-01	1080
c597d792e215	OS 2	Email Client 1	2021-09-01	1106
c603d749e374	OS 1	Email Client 1	2021-12-01	1028
c986d200e170	OS 2	Email Client 2	2021-09-01	1184
d168e758f876	OS 2	Email Client 1	2021-09-01	1107
d280e557f635	OS 3	Email Client 1	2021-03-01	0
d336e475f676	OS 2	Email Client 2	2021-06-01	1029
d394e816f943	OS 3	Email Client 2	2021-03-01	1003
d647e310f618	OS 2	Email Client 2	2021-06-01	1081

Purpose

Retrieved all available device information from the `machines` table to assess organizational assets requiring updates.

2. Retrieve Device ID and Email Client

SQL Query

SELECT device_id, email_client
FROM machines;

```
MariaDB [organization]> SELECT device_id, email_client FROM machines;
+--------------+----------------+
| device_id    | email_client   |
+--------------+----------------+
| a184b775c707 | Email Client 1 |
| a192b174c940 | Email Client 1 |
| a305b818c708 | Email Client 2 |
| a317b635c465 | Email Client 2 |
| a320b137c219 | Email Client 2 |
| a398b471c573 | Email Client 2 |
| a667b270c984 | Email Client 1 |
| a821b452c176 | Email Client 2 |
| a998b568c863 | Email Client 1 |
| b157c491d493 | Email Client 1 |
| b239c825d303 | Email Client 1 |
| b264c773d977 | Email Client 2 |
| b265c937d713 | Email Client 1 |
| b433c245d868 | Email Client 1 |
| b551c837d758 | Email Client 1 |
| b566c710d544 | Email Client 1 |
| b806c503d354 | Email Client 1 |
| b979c871d361 | Email Client 1 |
| c116d593e558 | Email Client 1 |
| c150d982e144 | Email Client 2 |
| c185d679e493 | Email Client 2 |
| c406d877e950 | Email Client 1 |
| c547d140e477 | Email Client 1 |
| c568d742e974 | Email Client 2 |
| c597d792e215 | Email Client 1 |
| c603d749e374 | Email Client 1 |
| c986d200e170 | Email Client 2 |
| d168e758f876 | Email Client 1 |
| d280e557f635 | Email Client 1 |
```

Purpose

Identified email clients installed on organizational devices for software review and update verification.

3. Retrieve Operating System and Patch Information

SQL Query

SELECT device_id, operating_system, OS_patch_date
FROM machines;

```
MariaDB [organization]> SELECT device_id, operating_system, OS_patch_date FROM machines;
+--------------+------------------+---------------+
| device_id    | operating_system | OS_patch_date |
+--------------+------------------+---------------+
| a184b775c707 | OS 1             | 2021-09-01    |
| a192b174c940 | OS 2             | 2021-06-01    |
| a305b818c708 | OS 3             | 2021-06-01    |
| a317b635c465 | OS 1             | 2021-03-01    |
| a320b137c219 | OS 2             | 2021-03-01    |
| a398b471c573 | OS 3             | 2021-12-01    |
| a667b270c984 | OS 1             | 2021-03-01    |
| a821b452c176 | OS 2             | 2021-12-01    |
| a998b568c863 | OS 3             | 2021-12-01    |
| b157c491d493 | OS 2             | 2021-03-01    |
| b239c825d303 | OS 1             | 2021-03-01    |
| b264c773d977 | OS 2             | 2021-03-01    |
| b265c937d713 | OS 2             | 2021-09-01    |
| b433c245d868 | OS 1             | 2021-06-01    |
| b551c837d758 | OS 3             | 2021-03-01    |
| b566c710d544 | OS 1             | 2021-06-01    |
| b806c503d354 | OS 2             | 2021-12-01    |
| b979c871d361 | OS 2             | 2021-03-01    |
| c116d593e558 | OS 3             | 2021-09-01    |
| c150d982e144 | OS 2             | 2021-06-01    |
| c185d679e493 | OS 1             | 2021-09-01    |
| c406d877e950 | OS 2             | 2021-06-01    |
| c547d140e477 | OS 2             | 2021-03-01    |
| c568d742e974 | OS 2             | 2021-09-01    |
| c597d792e215 | OS 2             | 2021-09-01    |
| c603d749e374 | OS 1             | 2021-12-01    |
| c986d200e170 | OS 2             | 2021-09-01    |
| d168e758f876 | OS 2             | 2021-09-01    |
| d280e557f635 | OS 3             | 2021-03-01    |
| d336e475f676 | OS 2             | 2021-06-01    |
```

Purpose

Reviewed operating systems and patch dates to determine whether devices required security updates.

Task 2 – Investigate Login Activity

1. Retrieve Login Attempt Locations

SQL Query

SELECT event_id, country
FROM log_in_attempts;

```
MariaDB [organization]> SELECT event_id, country FROM log_in_attempts;
+----------+---------+
| event_id | country |
+----------+---------+
|        1 | CAN     |
|        2 | CAN     |
|        3 | USA     |
|        4 | USA     |
|        5 | CANADA  |
|        6 | MEXICO  |
|        7 | CAN     |
|        8 | US      |
|        9 | MEX     |
|       10 | CANADA  |
|       11 | CANADA  |
|       12 | USA     |
|       13 | USA     |
|       14 | US      |
|       15 | USA     |
|       16 | CAN     |
|       17 | USA     |
|       18 | US      |
|       19 | US      |
|       20 | MEXICO  |
|       21 | US      |
|       22 | MEX     |
|       23 | MEXICO  |
|       24 | MEXICO  |
|       25 | US      |
|       26 | CANADA  |
|       27 | MEX     |
|       28 | MEXICO  |
|       29 | US      |
|       30 | MEX     |
```

Purpose

Investigated login attempt origins to identify connections from unexpected geographic locations.

2. Review Login Dates and Times

SQL Query

SELECT username, login_date, login_time
FROM log_in_attempts;

```
MariaDB [organization]> SELECT username, login_date, login_time FROM log_in_attempts;
+----------+------------+------------+
| username | login_date | login_time |
+----------+------------+------------+
| jrafael  | 2022-05-09 | 04:56:27   |
| apatel   | 2022-05-10 | 20:27:27   |
| dkot     | 2022-05-09 | 06:47:41   |
| dkot     | 2022-05-08 | 02:00:39   |
| jrafael  | 2022-05-11 | 03:05:59   |
| arutley  | 2022-05-12 | 17:00:59   |
| eraab    | 2022-05-11 | 01:45:14   |
| bisles   | 2022-05-08 | 01:30:17   |
| yappiah  | 2022-05-11 | 13:47:29   |
| jrafael  | 2022-05-12 | 09:33:19   |
| sgilmore | 2022-05-11 | 10:16:29   |
| dkot     | 2022-05-08 | 09:11:34   |
| mrah     | 2022-05-11 | 09:29:34   |
| sbaelish | 2022-05-10 | 10:20:18   |
| lyamamot | 2022-05-09 | 17:17:26   |
| mcouliba | 2022-05-11 | 06:44:22   |
| pwashing | 2022-05-11 | 02:33:02   |
| pwashing | 2022-05-11 | 19:28:50   |
| jhill    | 2022-05-12 | 13:09:04   |
| tshah    | 2022-05-12 | 18:56:36   |
| iuduike  | 2022-05-11 | 17:50:00   |
| rjensen  | 2022-05-11 | 00:59:26   |
| yappiah  | 2022-05-10 | 18:11:53   |
| arusso   | 2022-05-09 | 06:49:39   |
| sbaelish | 2022-05-09 | 07:04:02   |
| apatel   | 2022-05-08 | 17:27:00   |
| aalonso  | 2022-05-10 | 01:55:35   |
| aestrada | 2022-05-09 | 19:28:12   |
| bisles   | 2022-05-11 | 01:21:22   |
| yappiah  | 2022-05-09 | 03:22:22   |
```

Purpose

Analyzed login activity to determine whether authentication attempts occurred outside standard working hours.

3. Retrieve All Login Attempt Data

SQL Query

SELECT *
FROM log_in_attempts;

```
MariaDB [organization]> SELECT * FROM log_in_attempts;
+----------+----------+------------+------------+---------+-----------------+---------+
| event_id | username | login_date | login_time | country | ip_address      | success |
+----------+----------+------------+------------+---------+-----------------+---------+
|        1 | jrafael  | 2022-05-09 | 04:56:27   | CAN     | 192.168.243.140 |       1 |
|        2 | apatel   | 2022-05-10 | 20:27:27   | CAN     | 192.168.205.12  |       0 |
|        3 | dkot     | 2022-05-09 | 06:47:41   | USA     | 192.168.151.162 |       1 |
|        4 | dkot     | 2022-05-08 | 02:00:39   | USA     | 192.168.178.71  |       0 |
|        5 | jrafael  | 2022-05-11 | 03:05:59   | CANADA  | 192.168.86.232  |       0 |
|        6 | arutley  | 2022-05-12 | 17:00:59   | MEXICO  | 192.168.3.24    |       0 |
|        7 | eraab    | 2022-05-11 | 01:45:14   | CAN     | 192.168.170.243 |       1 |
|        8 | bisles   | 2022-05-08 | 01:30:17   | US      | 192.168.119.173 |       0 |
|        9 | yappiah  | 2022-05-11 | 13:47:29   | MEX     | 192.168.59.136  |       1 |
|       10 | jrafael  | 2022-05-12 | 09:33:19   | CANADA  | 192.168.228.221 |       0 |
|       11 | sgilmore | 2022-05-11 | 10:16:29   | CANADA  | 192.168.140.81  |       0 |
|       12 | dkot     | 2022-05-08 | 09:11:34   | USA     | 192.168.100.158 |       1 |
|       13 | mrah     | 2022-05-11 | 09:29:34   | USA     | 192.168.246.135 |       1 |
|       14 | sbaelish | 2022-05-10 | 10:20:18   | US      | 192.168.16.99   |       1 |
|       15 | lyamamot | 2022-05-09 | 17:17:26   | USA     | 192.168.183.51  |       0 |
|       16 | mcouliba | 2022-05-11 | 06:44:22   | CAN     | 192.168.172.189 |       1 |
|       17 | pwashing | 2022-05-11 | 02:33:02   | USA     | 192.168.81.89   |       1 |
|       18 | pwashing | 2022-05-11 | 19:28:50   | US      | 192.168.66.142  |       0 |
|       19 | jhill    | 2022-05-12 | 13:09:04   | US      | 192.168.142.245 |       1 |
|       20 | tshah    | 2022-05-12 | 18:56:36   | MEXICO  | 192.168.109.50  |       0 |
|       21 | iuduike  | 2022-05-11 | 17:50:00   | US      | 192.168.131.147 |       1 |
|       22 | rjensen  | 2022-05-11 | 00:59:26   | MEX     | 192.168.213.128 |       0 |
|       23 | yappiah  | 2022-05-10 | 18:11:53   | MEXICO  | 192.168.200.48  |       1 |
|       24 | arusso   | 2022-05-09 | 06:49:39   | MEXICO  | 192.168.171.192 |       1 |
|       25 | sbaelish | 2022-05-09 | 07:04:02   | US      | 192.168.33.137  |       1 |
|       26 | apatel   | 2022-05-08 | 17:27:00   | CANADA  | 192.168.123.105 |       1 |
|       27 | aalonso  | 2022-05-10 | 01:55:35   | MEX     | 192.168.103.210 |       0 |
|       28 | aestrada | 2022-05-09 | 19:28:12   | MEXICO  | 192.168.27.57   |       0 |
|       29 | bisles   | 2022-05-11 | 01:21:22   | US      | 192.168.85.186  |       0 |
```

Purpose

Obtained complete visibility into login events for further investigation and correlation.

Task 3 – Order Login Attempt Data

1. Sort Login Attempts by Date

SQL Query

SELECT *
FROM log_in_attempts
ORDER BY login_date;

```
MariaDB [organization]>  SELECT * FROM log_in_attempts ORDER BY login_date;
+----------+----------+------------+------------+---------+-----------------+---------+
| event_id | username | login_date | login_time | country | ip_address      | success |
+----------+----------+------------+------------+---------+-----------------+---------+
|      145 | ivelasco | 2022-05-08 | 09:06:02   | CANADA  | 192.168.39.196  |       1 |
|      163 | tmitchel | 2022-05-08 | 09:21:16   | MEX     | 192.168.119.29  |       0 |
|       36 | asundara | 2022-05-08 | 09:00:42   | US      | 192.168.78.151  |       1 |
|      165 | jreckley | 2022-05-08 | 15:28:43   | MEXICO  | 192.168.34.193  |       0 |
|      168 | jlansky  | 2022-05-08 | 13:25:42   | USA     | 192.168.210.94  |       1 |
|      169 | alevitsk | 2022-05-08 | 08:10:43   | CANADA  | 192.168.210.228 |       0 |
|       72 | alevitsk | 2022-05-08 | 12:09:10   | CANADA  | 192.168.139.176 |       1 |
|      101 | sbaelish | 2022-05-08 | 12:01:22   | US      | 192.168.145.158 |       0 |
|      172 | mabadi   | 2022-05-08 | 08:06:50   | US      | 192.168.180.41  |       1 |
|      150 | nmason   | 2022-05-08 | 14:40:02   | CAN     | 192.168.204.124 |       0 |
|       68 | mrah     | 2022-05-08 | 17:16:13   | US      | 192.168.42.248  |       1 |
|       66 | aestrada | 2022-05-08 | 21:58:32   | MEX     | 192.168.67.223  |       1 |
|       53 | nmason   | 2022-05-08 | 11:51:38   | CAN     | 192.168.133.188 |       1 |
|      147 | yappiah  | 2022-05-08 | 06:04:34   | MEX     | 192.168.65.245  |       0 |
|      148 | daquino  | 2022-05-08 | 06:15:55   | CANADA  | 192.168.135.6   |       1 |
|       49 | asundara | 2022-05-08 | 14:00:01   | US      | 192.168.173.213 |       0 |
|       47 | dkot     | 2022-05-08 | 05:06:45   | US      | 192.168.233.24  |       1 |
|       44 | daquino  | 2022-05-08 | 07:02:35   | CANADA  | 192.168.168.144 |       0 |
|       43 | mcouliba | 2022-05-08 | 02:35:34   | CANADA  | 192.168.16.208  |       0 |
|       56 | acook    | 2022-05-08 | 04:56:30   | CAN     | 192.168.209.130 |       1 |
|       80 | cjackson | 2022-05-08 | 02:18:10   | CANADA  | 192.168.33.140  |       1 |
|      117 | bsand    | 2022-05-08 | 00:19:11   | USA     | 192.168.197.187 |       0 |
|       12 | dkot     | 2022-05-08 | 09:11:34   | USA     | 192.168.100.158 |       1 |
|      189 | nmason   | 2022-05-08 | 05:37:24   | CANADA  | 192.168.168.117 |       1 |
|      191 | cjackson | 2022-05-08 | 06:46:07   | CANADA  | 192.168.7.187   |       0 |
|        8 | bisles   | 2022-05-08 | 01:30:17   | US      | 192.168.119.173 |       0 |
|      193 | lrodriqu | 2022-05-08 | 07:11:29   | US      | 192.168.125.240 |       0 |
|        4 | dkot     | 2022-05-08 | 02:00:39   | USA     | 192.168.178.71  |       0 |
|      197 | jsoto    | 2022-05-08 | 09:05:09   | US      | 192.168.36.21   |       0 |
```

Purpose

Organized login attempts chronologically to improve visibility into user activity trends.

2. Sort Login Attempts by Date and Time

SQL Query

SELECT *
FROM log_in_attempts
ORDER BY login_date, login_time;

```
MariaDB [organization]> SELECT * FROM log_in_attempts ORDER BY login_date, login_time;
+----------+----------+------------+------------+---------+-----------------+---------+
| event_id | username | login_date | login_time | country | ip_address      | success |
+----------+----------+------------+------------+---------+-----------------+---------+
|      117 | bsand    | 2022-05-08 | 00:19:11   | USA     | 192.168.197.187 |       0 |
|       92 | pwashing | 2022-05-08 | 00:36:12   | US      | 192.168.247.219 |       0 |
|        8 | bisles   | 2022-05-08 | 01:30:17   | US      | 192.168.119.173 |       0 |
|        4 | dkot     | 2022-05-08 | 02:00:39   | USA     | 192.168.178.71  |       0 |
|       80 | cjackson | 2022-05-08 | 02:18:10   | CANADA  | 192.168.33.140  |       1 |
|       43 | mcouliba | 2022-05-08 | 02:35:34   | CANADA  | 192.168.16.208  |       0 |
|      184 | alevitsk | 2022-05-08 | 03:09:48   | CAN     | 192.168.33.70   |       0 |
|       56 | acook    | 2022-05-08 | 04:56:30   | CAN     | 192.168.209.130 |       1 |
|       47 | dkot     | 2022-05-08 | 05:06:45   | US      | 192.168.233.24  |       1 |
|      189 | nmason   | 2022-05-08 | 05:37:24   | CANADA  | 192.168.168.117 |       1 |
|      147 | yappiah  | 2022-05-08 | 06:04:34   | MEX     | 192.168.65.245  |       0 |
|      148 | daquino  | 2022-05-08 | 06:15:55   | CANADA  | 192.168.135.6   |       1 |
|      191 | cjackson | 2022-05-08 | 06:46:07   | CANADA  | 192.168.7.187   |       0 |
|       44 | daquino  | 2022-05-08 | 07:02:35   | CANADA  | 192.168.168.144 |       0 |
|      193 | lrodriqu | 2022-05-08 | 07:11:29   | US      | 192.168.125.240 |       0 |
|      172 | mabadi   | 2022-05-08 | 08:06:50   | US      | 192.168.180.41  |       1 |
|       83 | lrodriqu | 2022-05-08 | 08:10:23   | USA     | 192.168.67.69   |       1 |
|      169 | alevitsk | 2022-05-08 | 08:10:43   | CANADA  | 192.168.210.228 |       0 |
|       36 | asundara | 2022-05-08 | 09:00:42   | US      | 192.168.78.151  |       1 |
|      197 | jsoto    | 2022-05-08 | 09:05:09   | US      | 192.168.36.21   |       0 |
|      145 | ivelasco | 2022-05-08 | 09:06:02   | CANADA  | 192.168.39.196  |       1 |
|       12 | dkot     | 2022-05-08 | 09:11:34   | USA     | 192.168.100.158 |       1 |
|      163 | tmitchel | 2022-05-08 | 09:21:16   | MEX     | 192.168.119.29  |       0 |
|       53 | nmason   | 2022-05-08 | 11:51:38   | CAN     | 192.168.133.188 |       1 |
|      101 | sbaelish | 2022-05-08 | 12:01:22   | US      | 192.168.145.158 |       0 |
|       72 | alevitsk | 2022-05-08 | 12:09:10   | CANADA  | 192.168.139.176 |       1 |
|      178 | sgilmore | 2022-05-08 | 12:27:22   | CAN     | 192.168.52.216  |       0 |
|      168 | jlansky  | 2022-05-08 | 13:25:42   | USA     | 192.168.210.94  |       1 |
|       49 | asundara | 2022-05-08 | 14:00:01   | US      | 192.168.173.213 |       0 |
|      150 | nmason   | 2022-05-08 | 14:40:02   | CAN     | 192.168.204.124 |       0 |
|      165 | jreckley | 2022-05-08 | 15:28:43   | MEXICO  | 192.168.34.193  |       0 |
|       68 | mrah     | 2022-05-08 | 17:16:13   | US      | 192.168.42.248  |       1 |
```

Purpose

Further refined event sequencing by sorting login attempts according to both date and time.

Investigation Summary

The SQL investigation revealed:

- Multiple organizational devices using different email clients
- Several systems with outdated operating system patch dates
- Login activity originating only from approved geographic regions
- Chronological ordering improved visibility into authentication patterns and user activity timelines

These findings demonstrate practical cybersecurity analysis using SQL for security monitoring and authentication investigations.

Skills Demonstrated

- SQL database querying
- Data retrieval and filtering
- Security log investigation
- Authentication activity analysis
- Database table analysis
- Sorting and organizing security data
- MariaDB command-line usage

Cybersecurity Relevance

Security analysts frequently use SQL to investigate authentication logs, identify suspicious activity, and monitor device security posture. Understanding how to query databases efficiently is essential for:

- Threat hunting
- Incident response
- Security monitoring
- Vulnerability management
- Compliance auditing

This lab strengthened practical experience in analyzing organizational security data through structured SQL queries.

Key Takeaways

- Learned how to retrieve specific and complete datasets from SQL databases
- Practiced analyzing authentication events for anomalies
- Used SQL sorting techniques to improve event investigation workflows
- Gained hands-on experience with MariaDB in a cybersecurity context

Conclusion

This lab provided practical experience in using SQL for cybersecurity investigations. By querying device and login activity data, I developed foundational skills required for security operations, threat analysis, and system monitoring. These techniques form the basis for more advanced database investigations and security analytics workflows.

Author: Cornelius Donkor